

# Introduction to Next Generation(Advanced) Distributed Systems Lecture

2025년도 2학기

# Contents

- Introduction to NGDS Lecture
- Cloud Foundation Trends from 2013
- AI Foundation Trends from 2018
- Cloud Computing Infra Market Trends
- 이론 수업 내용

# 대면 강의 시간 및 장소

- 화요일 9:00-11:00
  - 이론(대면: 신공 1214)
- 목요일 9:00-11:00
  - 실습 (대면: 신공 1214)

# 소통 및 상담

- 소통

- 카카오톡 오픈채팅방: DMS-Class-NGDS-2502
- <https://open.kakao.com/o/g92mqLPh>: 참여 코드 (2502)
- '이름(건대이메일ID)'로 들어오길 바람
- 건대 이메일 주소' 올리기
- '이름(건대이메일ID)'로 들어오길 바람

- Office Hour (개별 연락 시간 약속)

- Email: [dkmin@konkuk.ac.kr](mailto:dkmin@konkuk.ac.kr)

# 강의도구: MS Teams

- 사용용도
  - 강의 자료 공유
  - 강의 녹화 자료 공유
  - 프로젝트 팀 미팅공간 및 자료 공유
- 로그인
  - 건대 ID 로 MS Teams에 로그인한후 IAMDMS 도메인으로 변경하면 본강의 팀이름이 보임
- 팀이름
  - DMS-Class-NGDS-2502
- 참여 방법
  - 카톡 오픈채팅방에 자신의 건대이메일 주소를 올림
  - 조교가 IAMDMS 도메인에 건대 이메일로 등록해줄 것임
- 2번째 부터는 참여
  - 팀 이름을 선택하여 참가하면 됨

# 강의자료

- DMS-Class-NGDS-2502 팀의 파일 Tab
  - 이론
    - 파일/03-이론강의자료
    - eCampus 강의자료
  - 실습
    - 파일/04-실습강의자료
    - eCampus 강의자료
  - 1-Tip Review
    - 파일/05-1Tip Review
  - 녹화파일
    - 파일/Recordings

# 출석

- eCampus의 “스마트 출석” 사용
- 대면 참여한 본인만 출석인정

# 성적 (% 변경될 수 있음)

- 중간고사 40%
- 기말 텀 프로젝트 40%
- 과제물 15% (이론 8점, 실습 7점)
- 출석 5%
- 이론과제 8점
  - 중간고사 이전: 1-TipReview 3점 (참여도에 따라 상중하/3,2,1)
  - 중간고사 이후(5번 이론강의): 매 이론강의 1페이지 요약 (총 5점)

❖참고: 상대평가 과목임

# IT Context in 2013

2005



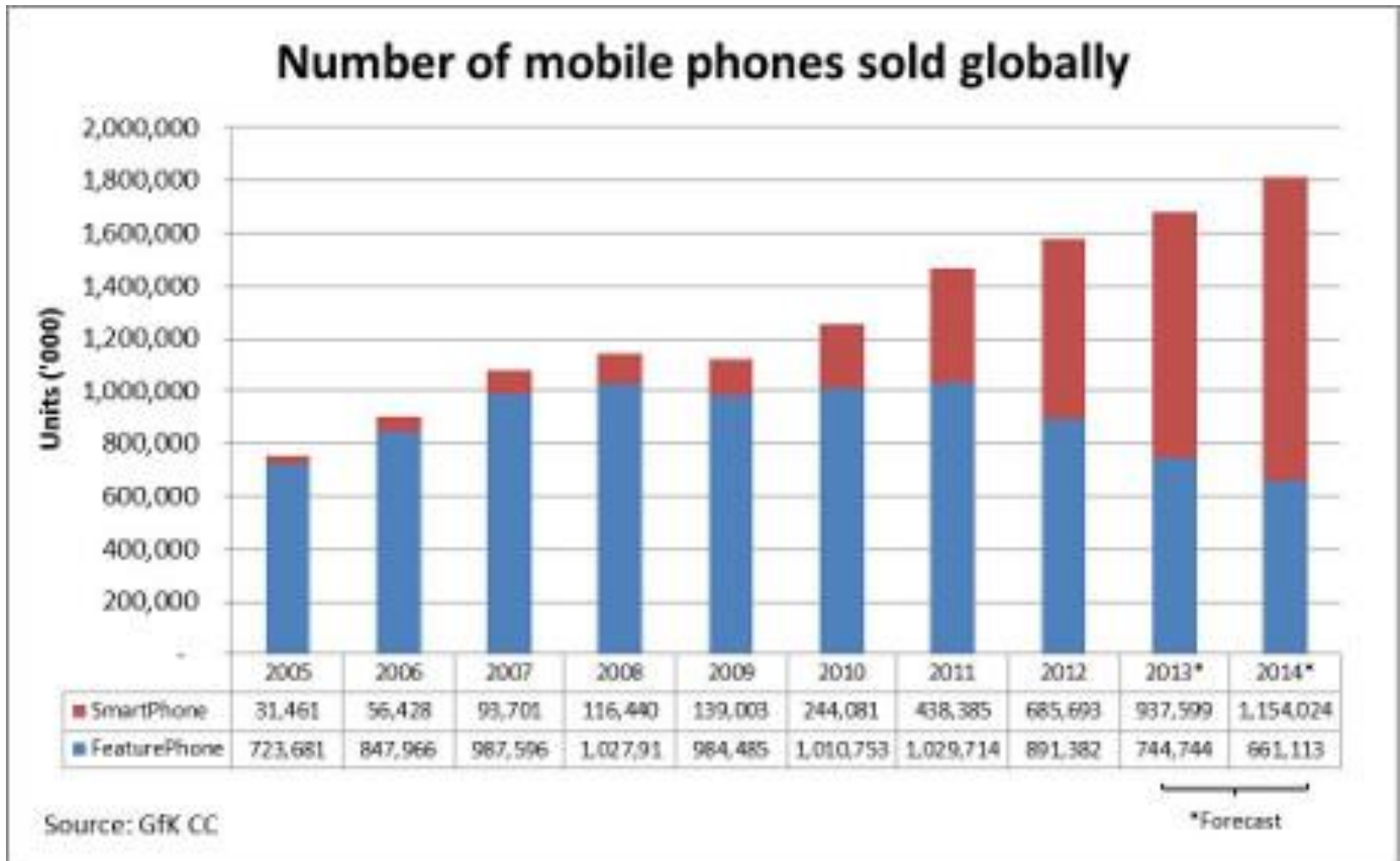
2013



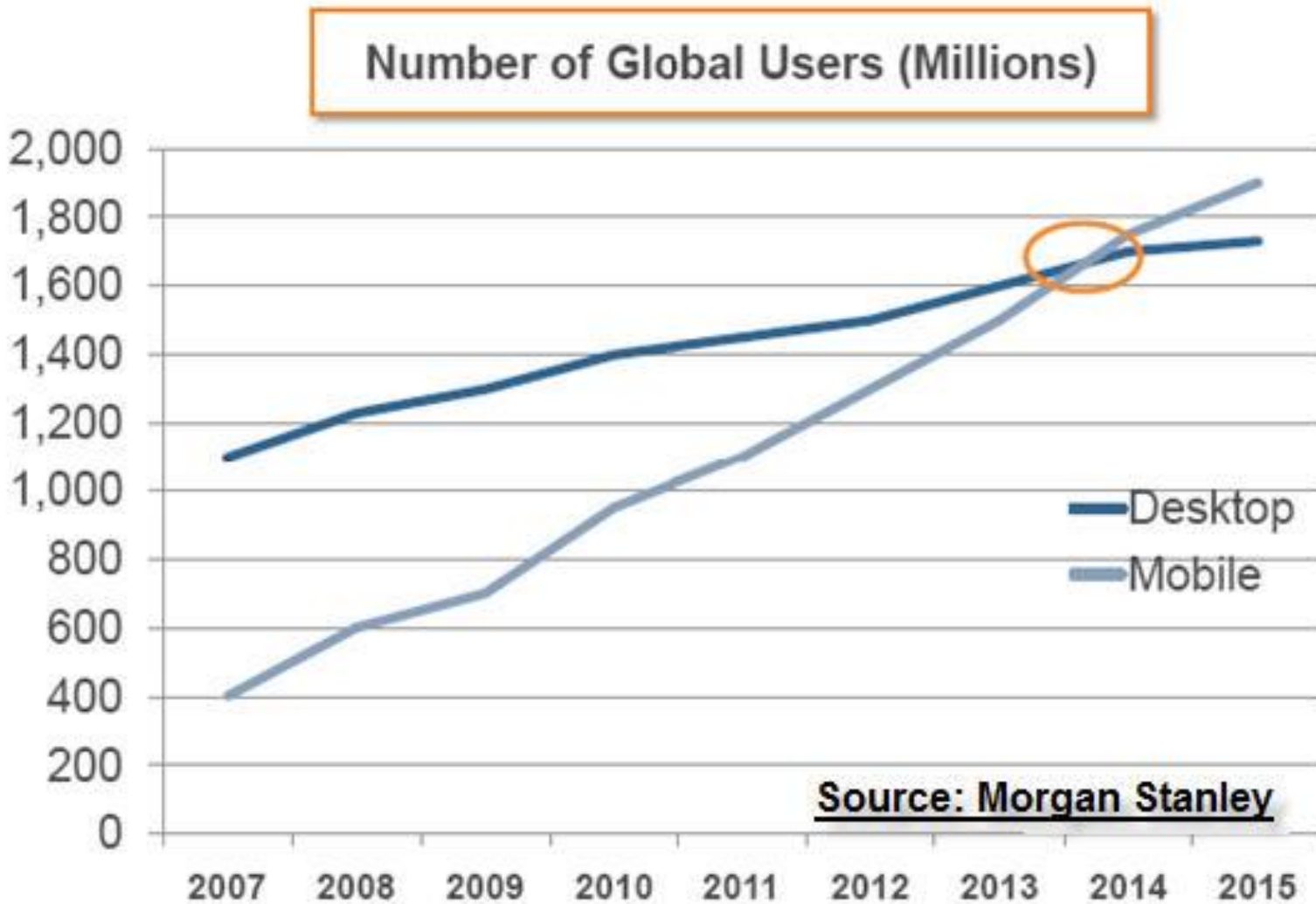
# Can you see the difference?



# Feature Phones vs. Smart Phones

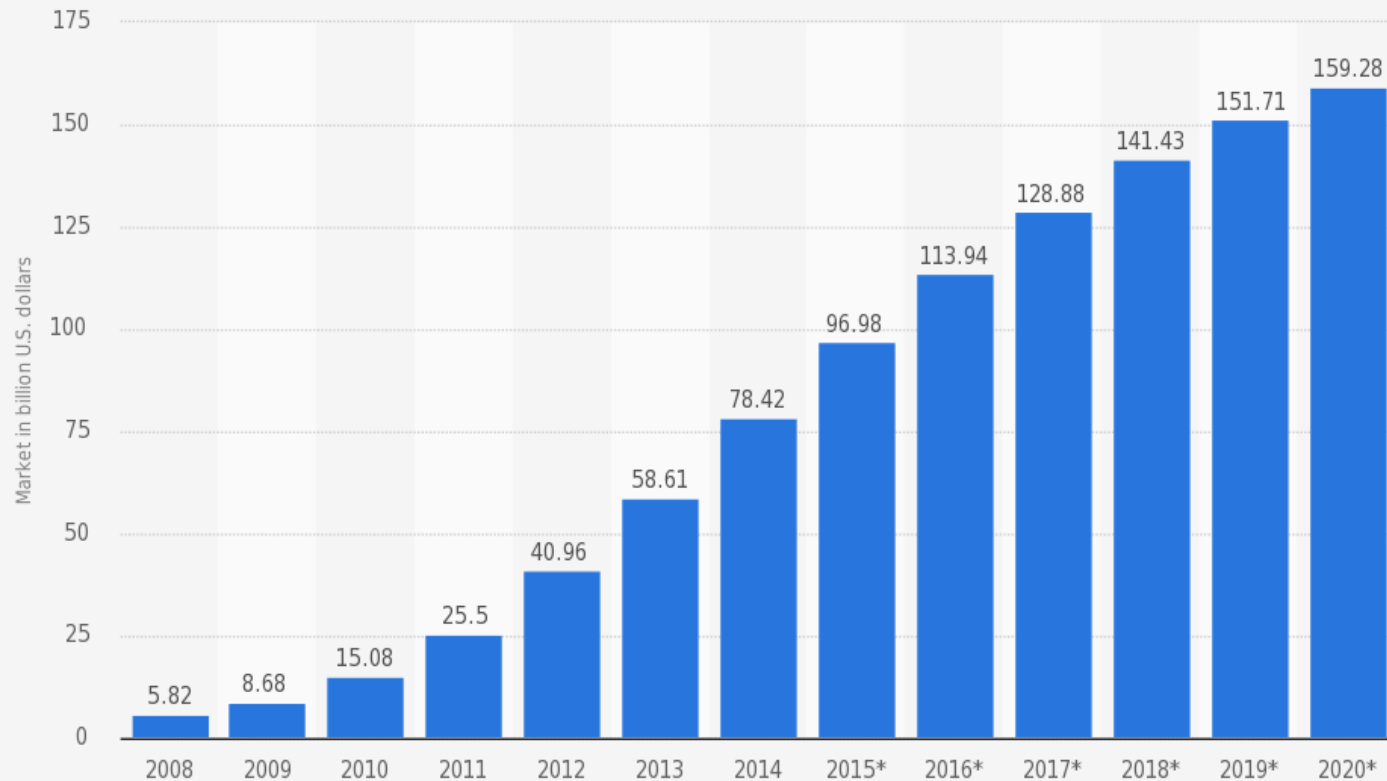


# PC vs. Smart Phone



# Cloud Market

**Total size of the public cloud computing market from 2008 to 2020 (in billion U.S. dollars)**



Source:  
© Statista 2016

Additional Information:  
Worldwide; 2008 to 2014

# Drivers for IT Innovation

- From 2013

## Mobile



Open for business  
any place, any time

## Social



Enables new levels of  
customer intimacy

## Big Data



New analytical  
opportunities at a  
massive scale

## Internet of Things



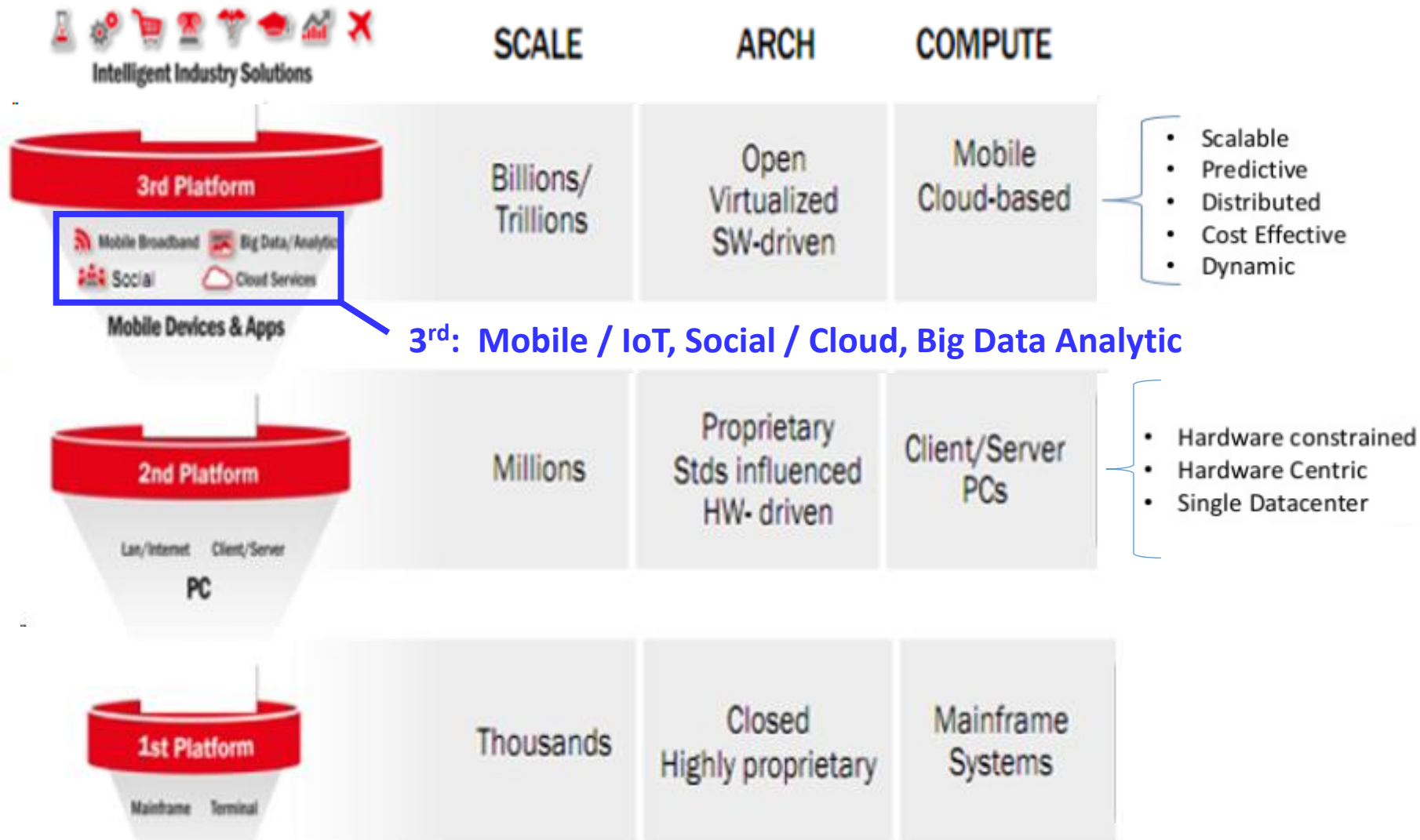
Explosive growth in  
Device to Device  
communication

## Cloud

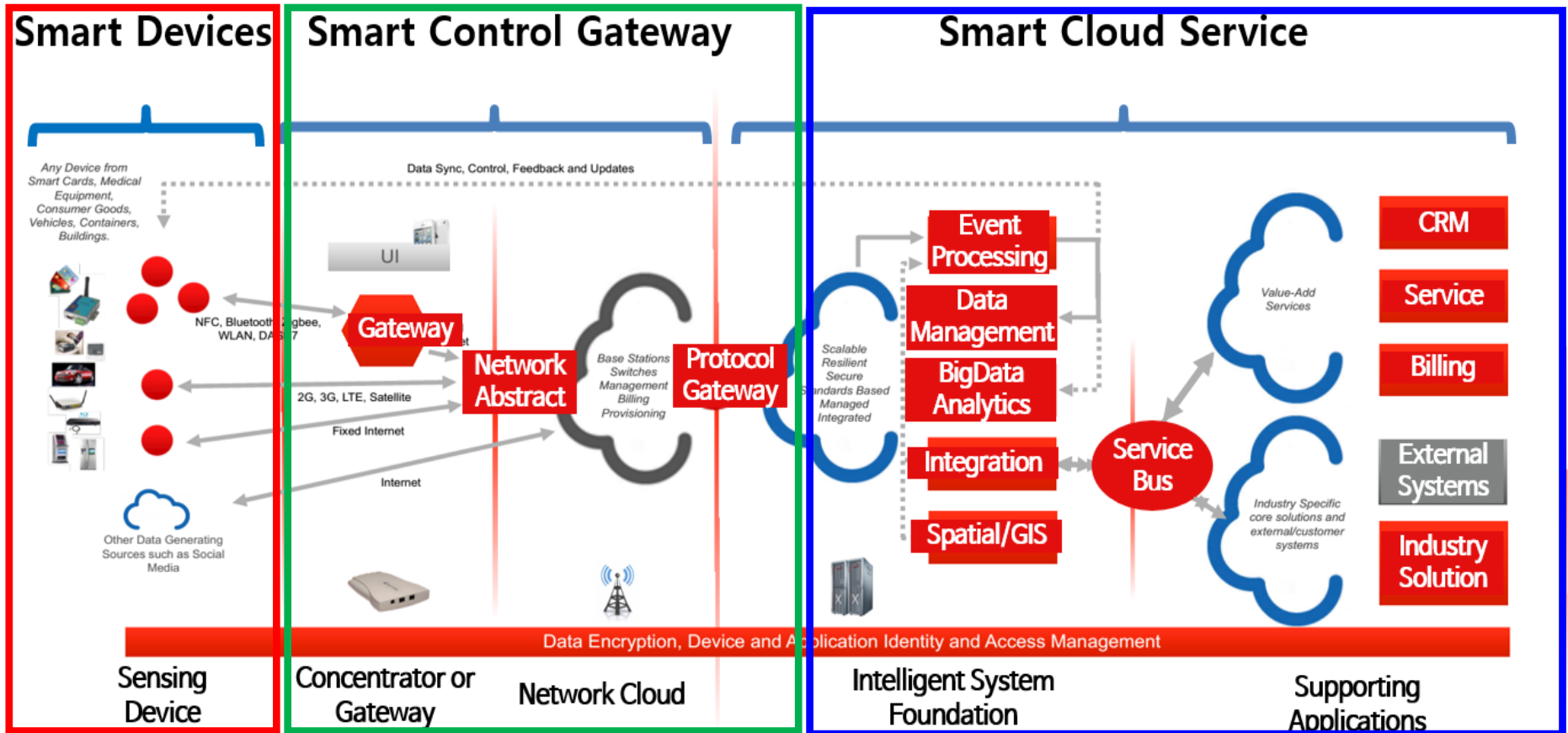


Foundation for efficiency  
and business agility

# Digital Transformation to 3<sup>rd</sup> Platform



# Device-to-Service Value Chain



# Top 10 IT Trends in 2018

## Intelligent



AI Foundation

Intelligent Apps  
and Analytics

Intelligent Things

## Digital



Digital Twin

Cloud to the Edge

Conversational Platforms

Immersive Experience

## Mesh



Blockchain

Event Driven Model

Continuous Adaptive

Risk and Trust

# AI & Advanced Machine Learning

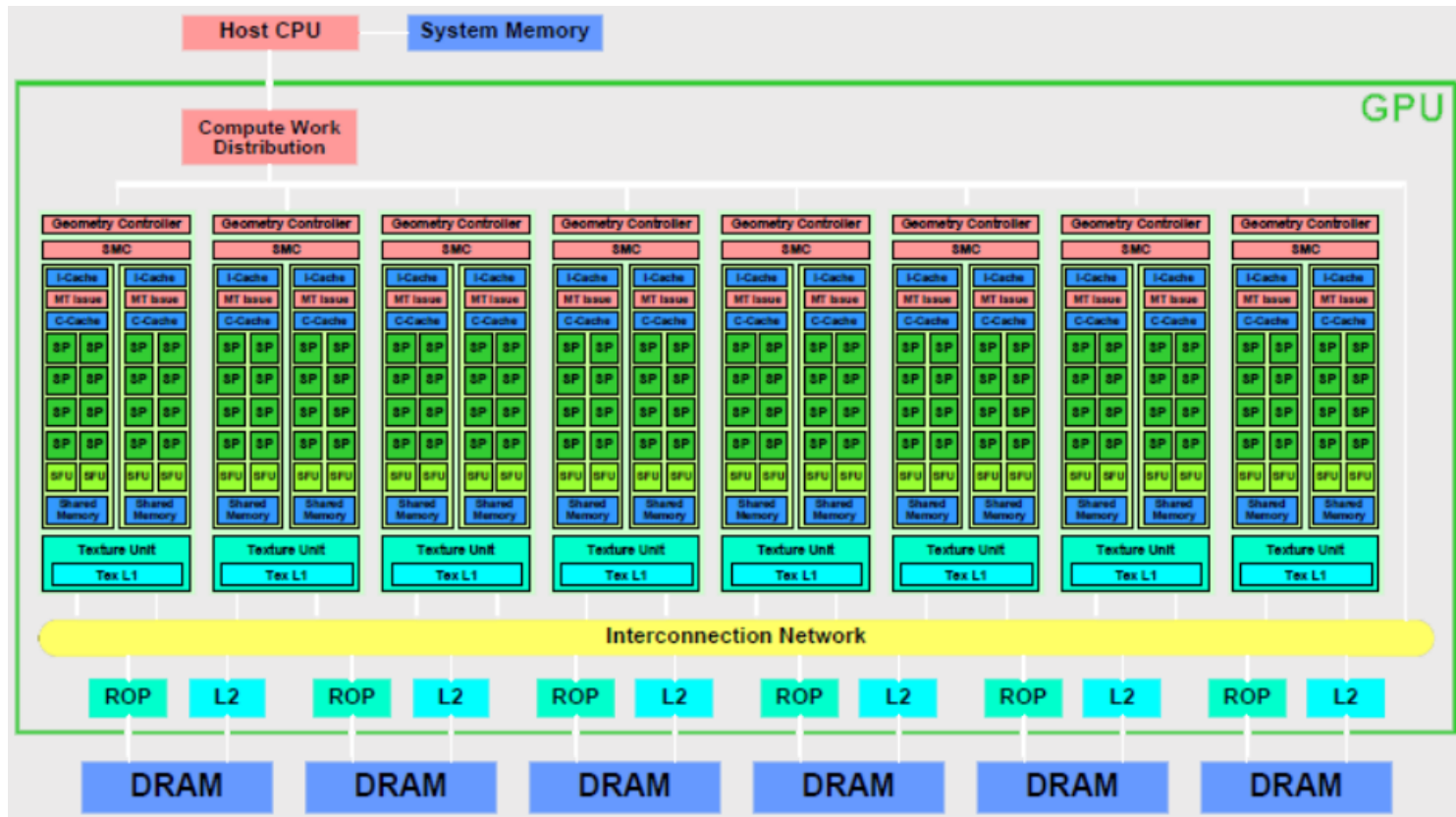
“This new era has been unleashed by the combination of

- extensive **parallel processing power**,
- **advanced algorithms (Deep Learning)** and
- **massive data sets** to feed the algorithms”

\* **unleashed**: (강력한 반응 • 감정 등을) 촉발시키다.

# Extensive Parallel Processing Power

- GPU



# Extensive Parallel Processing Power

- 8 GPU Server



# Extensive Parallel Processing Power

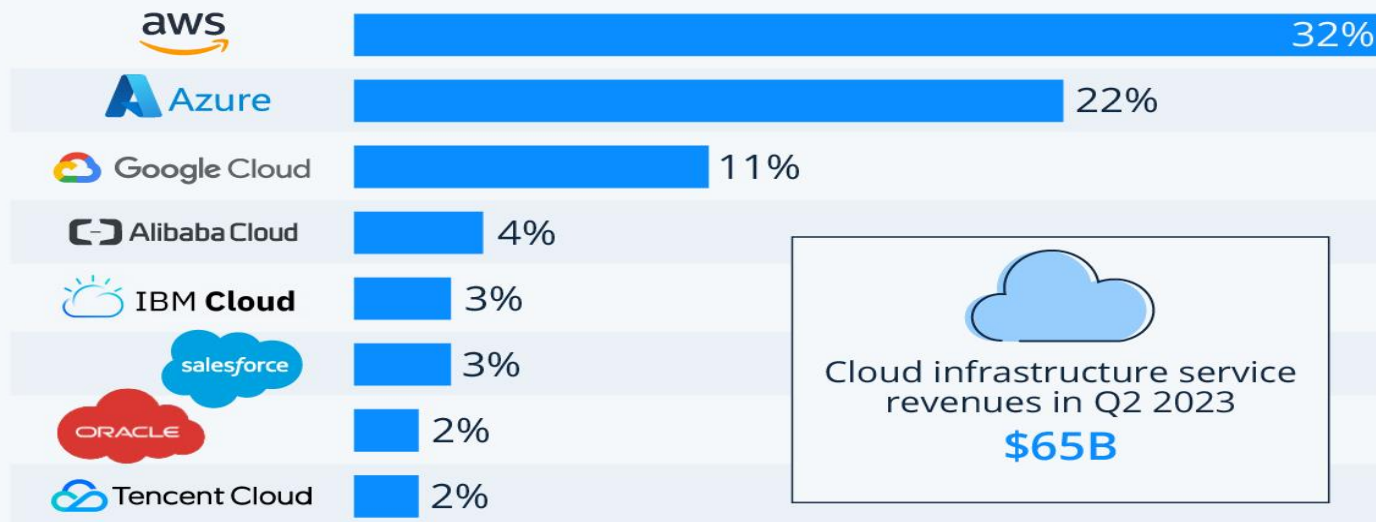
- **Cloud Computing** Clusters in Data Center



# Cloud Computing Infra Marketshare

## Amazon Maintains Lead in the Cloud Market

Worldwide market share of leading cloud infrastructure service providers in Q2 2023\*



\* Includes platform as a service (PaaS) and infrastructure as a service (IaaS) as well as hosted private cloud services

Source: Synergy Research Group



# Data Center Tour

- Inside Amazon's Massive Data Center
  - <https://url.kr/lcyb2d>
- Microsoft reveals its MASSIVE data center
  - [https://www.youtube.com/watch?v=80aK2\\_iwMOs](https://www.youtube.com/watch?v=80aK2_iwMOs)
- Google Data Center 360° Tour
  - <https://www.youtube.com/watch?v=zDAYZU4A3w0>

# Largest Data Centers

- Top 15 World Largest Data Center

| Data Center                   | Area (square ft.) | Country        |
|-------------------------------|-------------------|----------------|
| China Telecom Data Center     | 10 million        | China          |
| China Mobile                  | 7.2 million       | China          |
| The Citadel Campus            | 7.2 million       | United States  |
| Microsoft Data Center         | 7 million         | United States  |
| DuPont Fabros Technology      | 3.3 million       | United States  |
| CWL1 Data Centre              | 1.45 million      | United Kingdom |
| Apple's Mesa Data Center      | 1.3 million       | United States  |
| CoreSite Reston VA3           | 1.3 million       | United States  |
| Lakeside Technology Center    | 1.1 million       | United States  |
| Utah Data Center              | 1 million         | United States  |
| QTS Atlanta Metro Data Center | 0.97 million      | United States  |
| Tulip Data Center             | 0.9 million       | India          |
| Yotta NM1                     | 0.82 million      | India          |
| NAP of the Americas           | 0.75 million      | United States  |
| Next Generation Data Europe   | 0.75 million      | United Kingdom |

- Top 10 Largest Data Center Companies

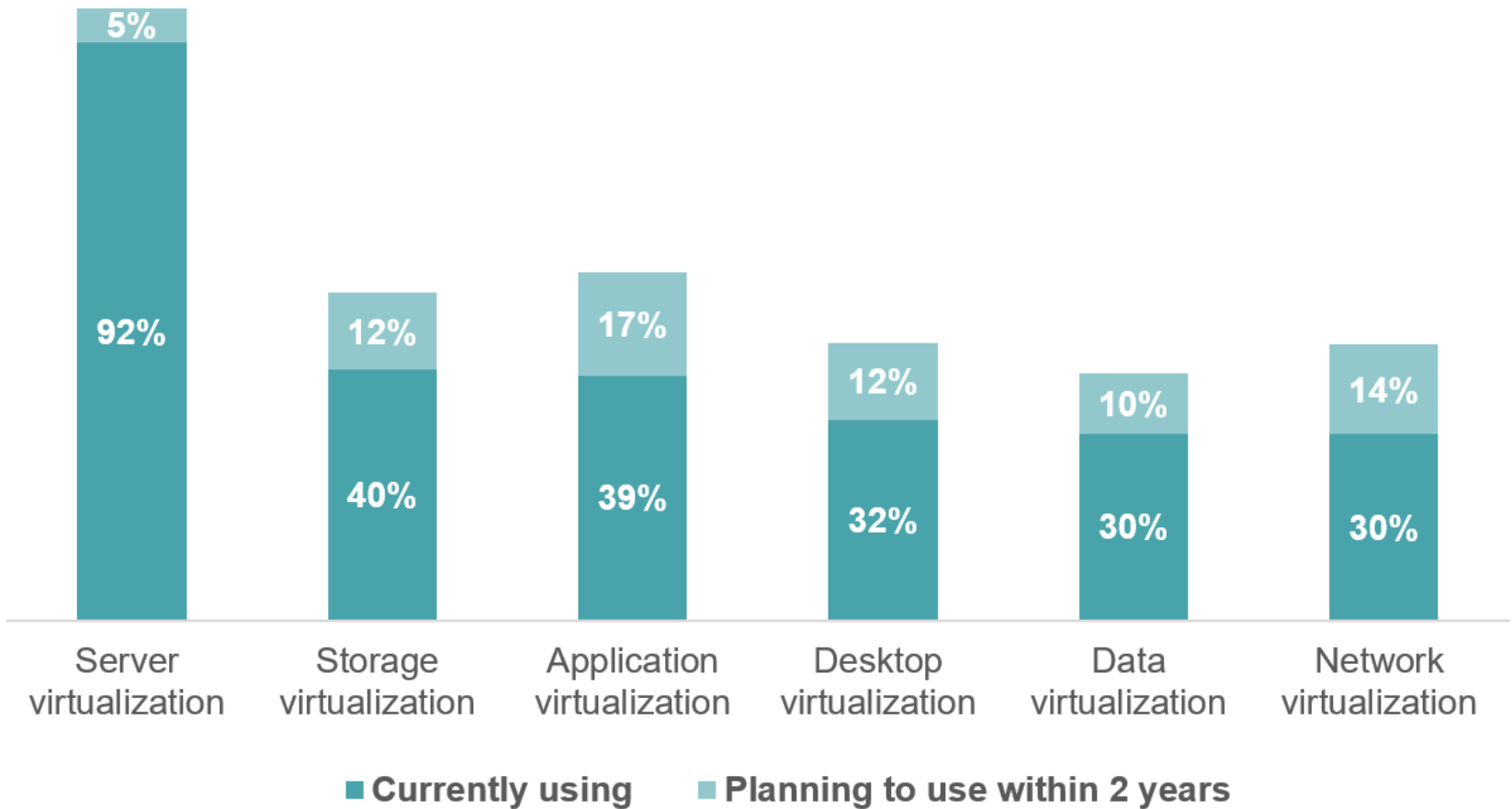
| Company                     | Annual Revenue  | Country |
|-----------------------------|-----------------|---------|
| China Unicom                | \$303.8 billion | China   |
| Amazon Web Services         | \$62 billion    | USA     |
| China Telecom               | \$55.3 billion  | China   |
| NTT Communications          | \$7.4 billion   | Japan   |
| Equinix                     | \$6.64 billion  | USA     |
| Digital Realty              | \$4.7 billion   | USA     |
| Cyxtera Technologies        | \$703.7 million | USA     |
| Coresite                    | \$638 million   | USA     |
| Switch (The Citadel Campus) | \$670 million   | USA     |
| 365 Data Centers            | \$21 million    | USA     |

# The 2020 State of Virtualization Technology

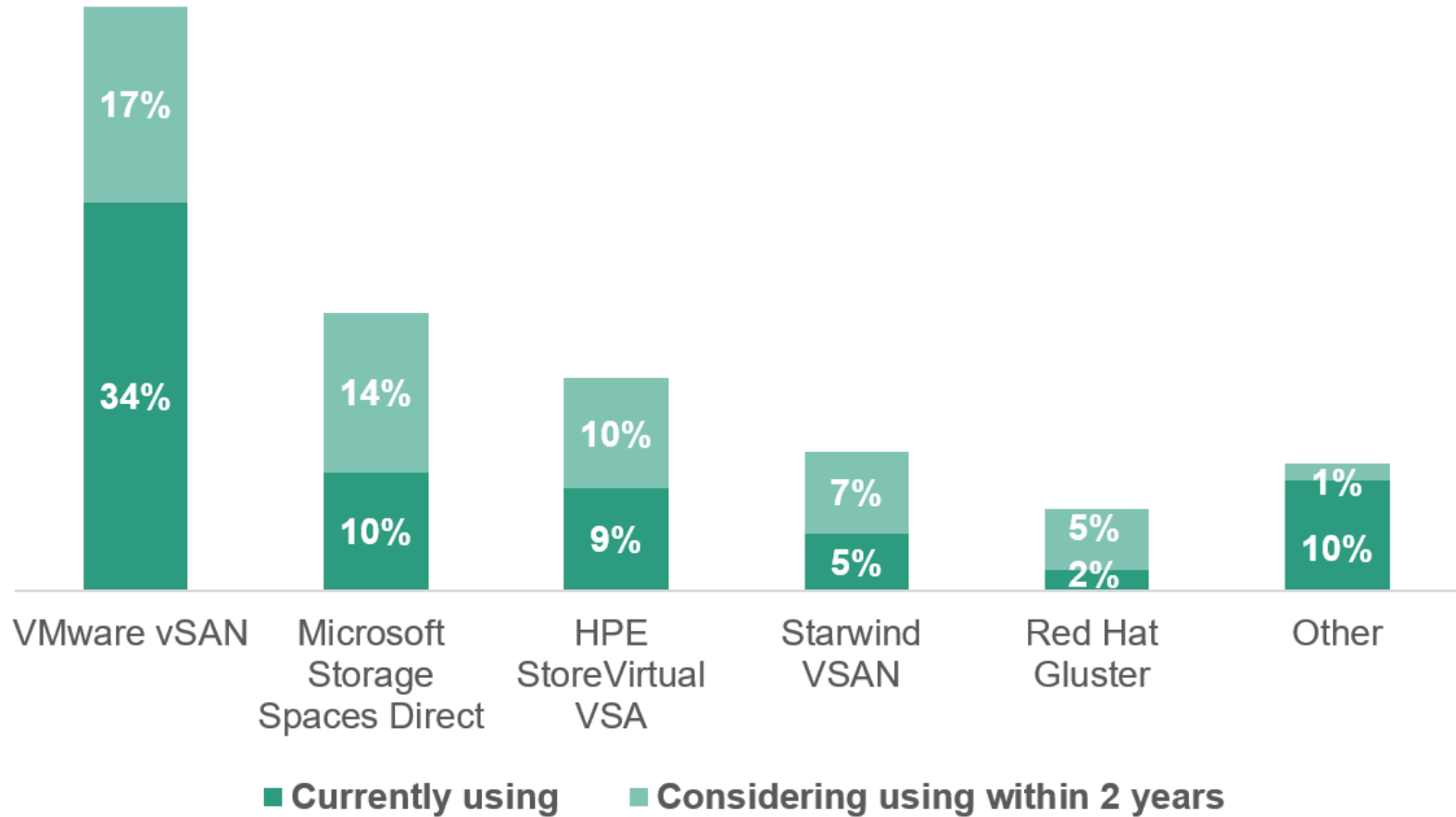
- Key Findings

- Virtualization adoption grows beyond servers
  - Within the next two years, double digit growth is expected in the use of desktop, application, network, storage, and data virtualization.
- A shift from storage arrays to software-defined storage
  - 25% of businesses plan to use software-defined storage solutions instead of buying shared storage arrays
- Virtual environments move to the cloud
  - Within the next two years, 45% of businesses are planning or considering moving their entire server virtualization environment to the cloud
- Server hypervisor technology viewed as a commodity
  - Nearly a third of IT decision makers believe server hypervisor technology is a commodity.

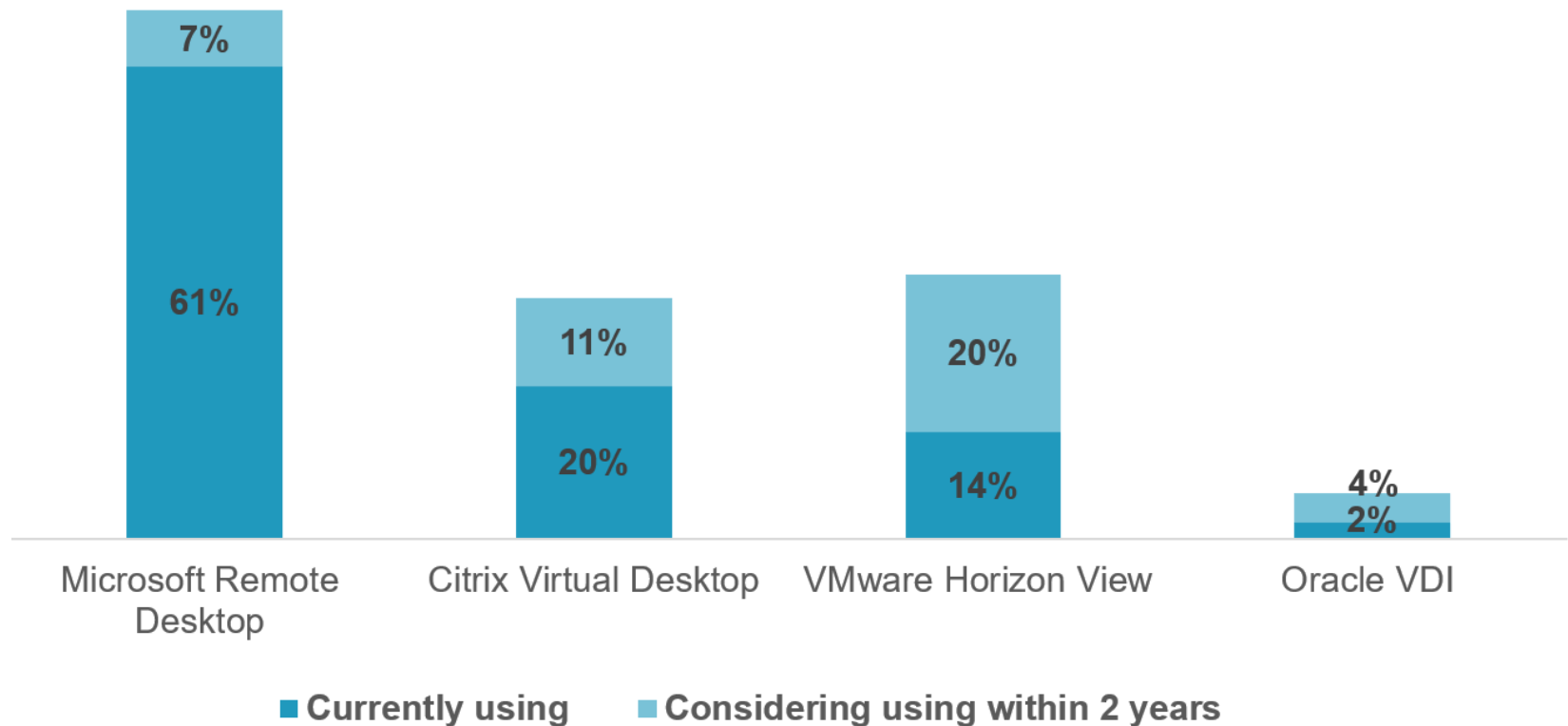
# Business Adoption of Virtualization Technology(2020)



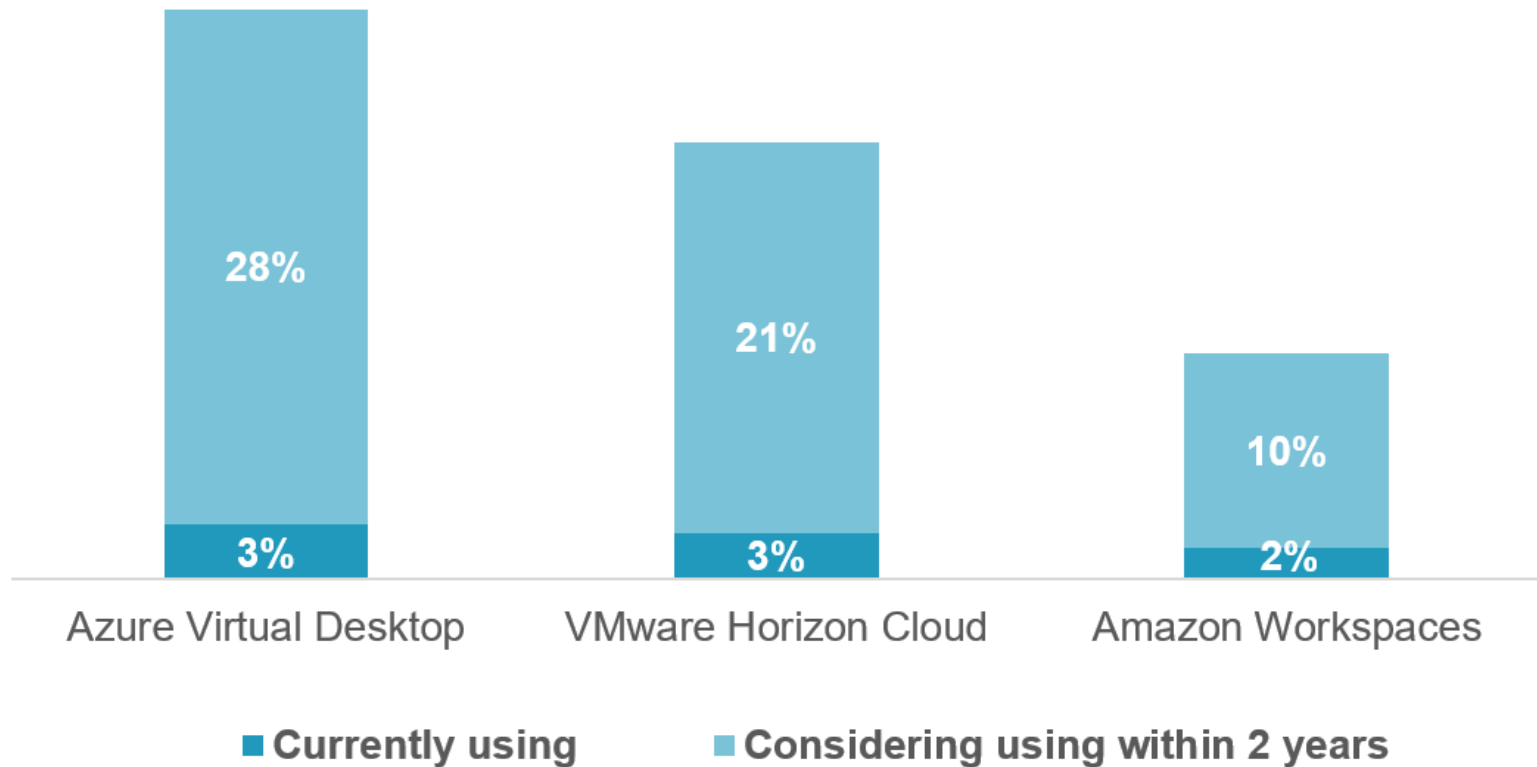
## Business Adoption of Storage Virtualization Solution(2020)



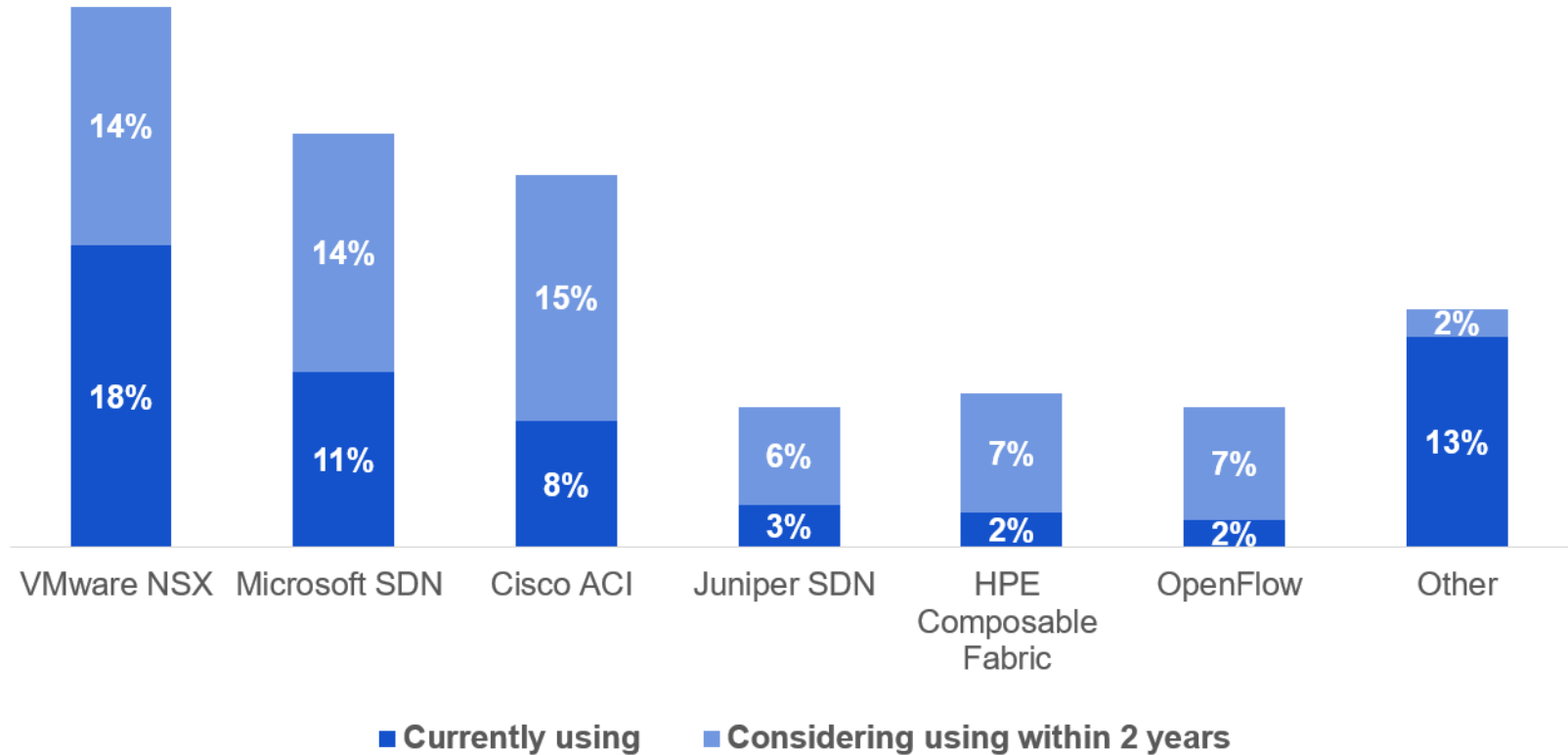
## Business Adoption of On-premises Desktop Virtualization Solutions(2020)



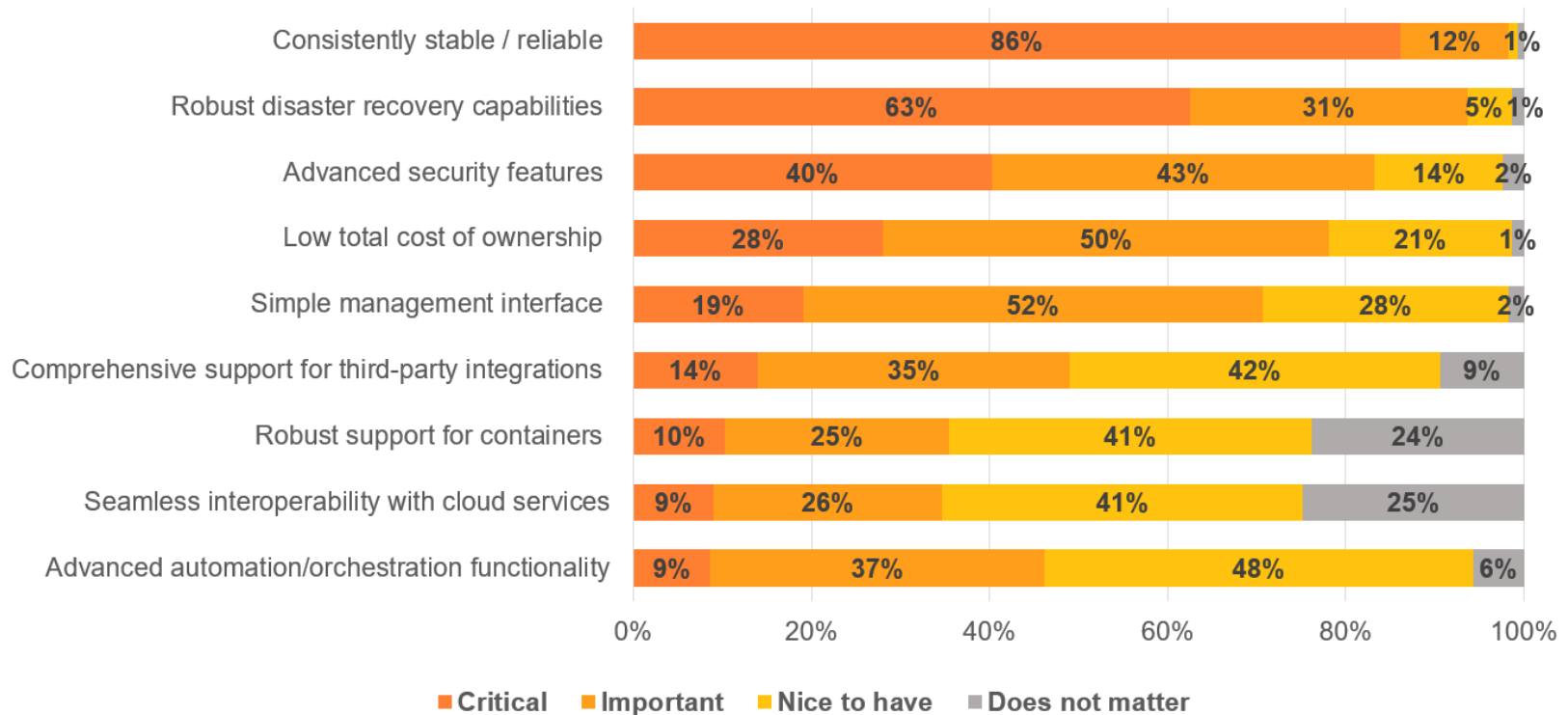
## Business Adoption of Cloud-Based Desktop Virtualization Solutions(2020)



# Business Adoption of Network Virtualization Solutions(2022)



# Importance of Consideration Factors When Evaluating Server Virtualization Solution (2020)



# 이론 수업 내용

- 1. NGDS 소개
- 2. Introduction to Docker
- 3. Docker Architecture and Technologies, 개인별 텀주제제안
- 4. Intro to Virtualization
- 5. Introduction to Kubernetes, 팀정하기
- 6. 휴강(추석)
- 7. Introduction to Cloud Computing
- 8. 중간고사
- 9. 제안서 발표
- 10. Cloud Computing Models and Services
- 11. Types of Hypervisor and Virtualization
- 12. Intro to Future Internet: SDN (Software Defined Network)
- 13. 중간발표
- 14. Intro to GPU Computing with CUDA
- 15. 2026 IT Trends by Gartner
- 16. 최종발표

# Summary

- Cloud Foundation Trends from 2013
  - to 3<sup>rd</sup> Platform from 2013
    - Open, Virtualized, SW-Driven
  - 5 Drivers for IT Innovation
    - Cloud, Mobile, Social, Big Data, IoT
- AI Foundation Trends from 2018
  - Distributed Parallel, Deep Learning Alg, Storage&Data
- Cloud Computing Infra Market Trends
  - The State of Virtualization Technology (2020)
- 이론 수업 내용
  - Cloud Computing, Virtualization, Orchestration
  - SW-Defined Network, GPU

# 1 Tip Review

- 당일 강의 요약틀 보고 강의 핵심 떠올려 보기
- 1 Tip Review 작성
  - 강의 관련 새롭게 배운 것 정리하기
  - 이해 안가는 것, 궁금한 것에 대해 질문 올리기
- 수업 끝나고 바로 하기

Thank You!!

Any Questions ??